HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained accurate **Revit** and **3D models** of industrial equipment and customer facilities, integrating mechanical, electrical, and controls information.
- Created and managed **Revit families** and reusable digital component libraries for efficient model development across multiple projects.
- Completed **3D scanning** utilising **Trimble X9 Scanner** to capture existing site conditions for detailed facility modelling.
- Prepared engineering models for use in **real-time environments**, optimising geometry and textures for performance in platforms like Unity and Unreal Engine.
- Supported the development of **simulation models** for automated manufacturing systems using **Visual Components**, detailing equipment movement and material flow.
- Explored the connection of simulated equipment data with 3D environments to support **digital twin** applications, visualising equipment status and process information.
- Containerised simulation and visualisation applications using **Docker** for consistent deployment across development and testing environments.

SKILLS

BIM & 3D Modelling – Autodesk Revit, Revit Families, 3D Scanning, Trimble X9 Scanner, Navisworks, Autodesk Construction Cloud

Simulation & Digital Twins – Visual Components, Emulate3D, Unity, Unreal Engine, Real-time 3D, Digital Twin

3D Asset Development – Blender, Asset Optimisation, Real-time Environments

Engineering Integration – Mechanical, Electrical, Controls, Automation Information, Engineering Drawings, Technical Information

Core Engineering Skills – Problem Solving, Independent Work, Communication, Multidisciplinary Environments

Development & Automation – Python, Docker

PROJECTS

Digital Twin Development for Factory Automation

- Developed a **digital twin** representation of an automated assembly line using **Unity**, integrating real-time sensor data feeds to visualise equipment status and material flow.
- Modeled complex robotic arm movements and queueing systems in **Visual Components** to simulate cycle times and identify potential bottlenecks, reducing expected physical testing by approximately 15%.
- Created optimised **3D assets** in Blender for the factory environment, ensuring high visual fidelity while maintaining real-time performance.

Revit-based BIM Content Creation for Industrial Equipment

- Engineered a library of **Revit families** for standard automation components, including PLCs, actuators, and sensors, improving consistency and reducing manual modelling time by ~20% per equipment set.
- Integrated detailed mechanical and electrical data into the **BIM models**, ensuring accuracy for downstream coordination and simulation tasks.
- Collaborated with automation engineers to refine model data structures and ensure compliance with project-specific BIM execution plans.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)